

Evasion Along Borders: How Trade Variables Affect the Size of the Shadow Economy

Alexander Minasyan

April 1, 2026

1 Introduction

The study of the shadow economy of countries is quite extensive, yet consistent results are often-times difficult to come by. There have been a number of determinants of shadow-economic size found, yet policy solutions are quite involved and difficult to put into practice. The shadow economy has a number of important policy implications and consequences. It is important to note that this phenomenon has been growing as the sizes of the shadow economies of all of the GNP nations have grown significantly in the period of 1960 - 1995 (Schneider and Enste, 2000). With an average percentage point increase of 10.4 (Schneider and Enste, 2000) and an average percentage increase of 486%, the issue of shadow economic size has been a growing consideration among researchers in the field of economics. Another startling statistic is that, according to the estimates of Schneider and Medina (2017), the largest shadow economies have their sizes estimated at around 63% of GDP (Bolivia, Guatemala, and Nigeria). This makes clear that shadow activity is a pervasive phenomenon and much must be done to address this issue. The main consequences of shadow activity can be summarized with the following.

To start, businesses that operate part of or all of their business in the shadow economy have the advantage in that they avoid paying taxes and regulatory fees (Tanzi, 2002). They can also skirt many of the regulations that “official” businesses are beholden to (Tanzi, 2002). This leads to unfair competition and can further lead businesses into the shadow economy as they try to even the playing field.

Adding to this, it is quite easy to see how a larger shadow economy would lead to a reduced tax base, thus putting a strain on social services. To compound this problem, many studies have found that the potentially heavy burden of taxes and especially social security contributions cause individuals to enter the shadow economy (Schneider and Enste, 2000). This creates a vicious cycle in which the presence of an abnormally large shadow economy causes a reduction in tax revenues, which causes a growth in the size of the shadow economy, which then puts a greater strain on the tax base and so on. It is likely that mitigating the size of the shadow economy will lead to a reduced tax burden, higher quality public services, and higher equality in taxation. We can also see the reverse of the cycle outlined above as greater incentive to tackle the issue of the shadow economy.

Additionally, increased shadow economic activity can discourage foreign direct investment (FDI) (Tanzi, 2002), particularly Greenfield investments (a component of FDI characterized by a company constructing operations without the purchase of partnership of local businesses) (Yusuf and Nguyen, 2023). This, in part, likely leads to a reduction in GDP growth that is often larger than the growth rate of a country’s shadow economy

(Loayza, 1996). Not only does the shadow economy have an effect on economic variables, it also distorts estimations of them. Since this activity is productive, laborious, revenue- and output-producing, and taxable, it is factored in a country's theoretical national income statistics, but since these activities cannot be reliably estimated, the national income statistics of nearly all countries are technically inaccurate. The higher the variability and uncertainty in these measures (note that they are quite high as illustrated by Schneider and Medina (2018)), the less governments and people can rely on these as accurate representations of their society, which can obviously lead to poor policy selection among other things.

Despite the wealth of research on the topic in general, research into the relationship of trade and the shadow economy is quite fringe and emblematic of mixed and uncertain results. Various papers have found results that are either significant or insignificant and either positive or negative. There has been no clear consensus on even the direction of this change, let alone the direction with a stable idea of significance. Theoretical implications tell us that an increase in trade liberalization will lead to increased competition, which will push producers into engaging in the shadow economy (mainly through labor as described by Fiess and Fugazza (2012)) for an edge. Additional theoretical work has shown that openness to trade leads to higher informal employment and higher underground wages (Kar and Marjit, 2001)(Marjit and Acharyya, 2003). Informal employment does tell the whole story, but is a substantial part of the shadow economy and is a major motivator of its size and growth. Among empirical works, a variety of different results have been found. With similar frameworks and control variables, a clear determination of the relationship between trade and the size of the shadow economy is ambiguous.

To remedy this, a more in-depth econometric model needs to be utilized and the relationship needs to be examined in a larger variety of ways. To start, this study uses a set of control variables that is more in-line with the seminal literature on the shadow economy in general. To truly isolate the effects of trade, it is crucial to choose a more comprehensive set of control variables. Additionally, current literature lacks diversity in the model selection. Without the presence of growth of the size of the shadow economy, certain nonlinearities, and interaction terms, it cannot be hoped to obtain a fundamental understanding of the relationship under study. Along with the preceding, certain variables that may better capture the effects of the certain types of underground activity are lacking in the literature and an analysis of these will also be present.

This paper will utilize the system of national accounts method to try to capture all parts of the shadow economy. Despite this less prevalent convention, it seems to be the most appropriate dataset instead of the most used set of data by Schneider and others. The reasons for its usage will be outlined in more detail later. To also capture this relationship on a global scale, panel data models will be used as well as a sample of many countries across a number of years. A large breakdown will be done starting with purely linear models (to better reflect the current literature); then adding a number of control variables; and finally incorporating alterations to standard models such as interactions terms, growth, and nonlinearities.

2 Literature Review

2.1 Understanding the Shadow Economy

One may have noticed that, thus far, the term "shadow economy" has exclusively been used. A major issue in the study of shadow economics is that there is no globally accepted definition of it. There are a number of alternative definitions and these can be encapsulated in different terms such as black economy, dark economy, informal economy / sector, underground economy, and others. For this paper, all of the above and others not stated (but relevant) will be used synonymously. There are also various axes in which shadow economic activity can be defined, namely monetary / nonmonetary transactions and legal / illegal activities. Different estimates utilize different parts of this 2-dimensional scheme. Table 1 below, by Schneider and Enste (2000), presents a good summary of the various components that can be included in the definition of the informal economy.

	Monetary Transaction		Nonmonetary Transactions	
Illegal Activities	Trade in stolen goods, drug dealing and manufacturing, prostitution, gambling, smuggling, and fraud.		Barter: drugs, stolen goods, smuggling, etc. Produce or growing of drugs for own use.	
	Tax Evasion	Tax Avoidance	Tax Evasion	Tax Avoidance
Legal Activities	Unreported income from self-employment	Employee discounts	Barter of legal services and goods	All do-it-yourself work and neighbor help
	Wages, salaries and assets from unreported work related to legal services and goods	Fringe benefits.		

Figure 1: A Taxonomy of Underground Economic Activities (Schneider and Enste, 2000)

To encapsulate all of the aspects that will be studied in this paper, the most comprehensive definition will be used. All parts of the above table will be included under the definition of the shadow economy and all of its synonyms. This will motivate the selection of our datasets and variables to better capture all effects and implications.

2.2 Main Causes of the Shadow Economy

Among the most cited reasons for shadow economic activity are tax-related issues, the regulatory environment, and the strength of institutions. To start, the often most cited reason for the growth of the shadow economy is the prevalence of taxes. Namely, businesses who want to avoid paying taxes will transfer some of their activities into the shadow economy where they can, through tax evasion and tax avoidance, save money on tax payments. This is mainly from the business perspective, yet one can localize this to individuals, wherein, the burden of taxes can cause individuals to move to the informal economy. This happens,

oftentimes, for two reasons. First of all, income taxes play a large role in this phenomenon as many, feeling the effects of a large income tax rate (i.e. a reduction in wages), will try to find work underground, where “wages” are higher (Schneider and Enste, 2000). And since this work is paid in cash and the activity illegal, the employer will not levy the tax on the individual. This is further supported by the findings of Schneider (1994) that direct taxes (social security contributions included) have the largest impact on the size of the shadow economy.

Additionally, large social security burdens will cause people to try to find a way out of them. This can realistically be done only one way, by losing your employment. When individuals are unemployed, they are not expected to pay these contributions, so they will move their labor to the shadow economy. This way, they can get paid, collect public-funded benefits, and not contribute to the society’s social safety net. It is clear why this cycle can create major issues and is briefly discussed in the introduction. As was noted in the introduction, the presence of a larger shadow economy can create a cycle, wherein, the tax base is reduced, leading to a strained social services system, and thus more SE activity, rinse and repeat.

Another important factor related to taxation is the strength of the tax-collecting institutions. Although institutions are an issue of their own, it is worth mentioning this. It was found in a variety of studies, summarized by Schneider and Enste (2000) and Tanzi (2002), that a weaker tax-collection system emboldens individuals to act in the informal sector. Tanzi (2002) describes this as a “cold war” between tax agencies and those who are in / willing to supplement above ground activity with underground activity. The higher the likelihood and the greater the punishment for evading / avoiding taxes, the less individuals will want to risk entering the shadow economy. This operates as an incentive structure where people, as they often do in other things, will weigh the risks and rewards of engaging in illicit activities. Similarly, Cebula (1997) finds that more IRS audits and higher penalties imposed for mis-representing due tax payments, decreases the size of the shadow economy. Another component of institutions is the tax laws themselves, specifically the ability to legally avoid taxes. It was found by Neck, Uwe Wächter, and Schneider (2012) that the more complex a tax system (i.e. the more wiggle room for avoidance) the lower the shadow economic activity a country will have in aggregate.

The second major driver of shadow economic activity is the prevalence of regulation. It is not difficult to see that an increase in regulation, similar to taxes, will cause people to shift towards the shadow economy to avoid it. The significance of this occurs in that increased labor regulations can increase costs, which once passed onto employees, can cause them to move into the underground economy with incentives similar to that of the taxation example above (Schneider and Pöll, 1999). Tanzi (2002 and 1983) supports this view, finding that with an overabundance of regulation comes a larger share of shadow economic activity. It is also important to note that it isn’t just the amount of regulation, but the quality of its enforcement. This is mainly related to institutions, but has its significance in regulations specifically.

The third reason that economists often cite as an explanation for shadow economic activity is the quality of institutions and, by extension, corruption. As was noted earlier, the strength of tax and regulatory institutions adds a layer of nuance to their respective determinants. Although tax and regulation law are major factors, the ability of institutions to enforce these provisions is considerable. From this, we see a similar cycle to the one outlined above, weak institutions create great incentives to go towards the informal econ-

omy, which then starts our destructive cycle. A strong rule of law and proper integration of property rights improves (i.e. reduces) the situation of a country regarding its shadow economy (Schneider and Buehn, 2017). It is not difficult to see why this happens and, once again, mirrors our discussion of taxes and regulations. The stronger the system of laws, the less likely people will be to risk engaging in shadow economic activity.

Also, an economy plagued with inefficiencies caused by poor institutions is more likely to see businesses move towards the shadow economy, as operating above ground is not efficient (Schneider and Buehn, 2017). To support this view, Cahn, Schinckus, and Thahn (2021) find, after controlling for a number of determinants of the shadow economy, that estimates from the WDI for corruption, government effectiveness, political stability and absence of violence / terrorism, regulatory quality, rule of law, and voice and accountability all have significantly negative effects on the size of the shadow economy. Interestingly enough, the effect of corruption on the shadow economy is quite mixed. It is determined by the wealth of the country under study (Dreher and Schneider, 2010). They showed that low-income countries have corruption and their shadow economic apparatus as compliments while they are not in high income countries. On the other hand, Dreher, Kotsogiannis, and McCorriston (2008) find that the shadow economy and corruption are, in general, substitutes.

2.3 Trade and the Shadow Economy

With the background above, it comes time to discuss the possible significance of trade and the size of the underground economy. The coming discussion will provide some theoretical reasons for the existence of this relation while also laying the foundations for the choice of variables that will be used. We start by defining what is meant by trade as this will motivate the choice of variables used. Trade will be defined simply along 3 dimensions: trade volume (e.g. imports or exports), trade barriers (e.g. taxes and regulations), and the quality of goods.

To begin our discussion, it is quite well-known that an increase in exports will, in the short term, increase jobs in exporting industries. Not only this, but comparative advantage theory states that, with increased comparative advantage in exports comes greater benefits to trade and thus quality of life (Sirgy et al., 2006). An improvement in the quality of life will lead to a reduced necessity of consumers to engage with the shadow economy. The factors that drive this are multiple. It would not be difficult to see why an increase in a country's exports (in the short term) lead to increases in jobs (thus a reduced incentive for job seekers to enter the informal economy) and sales revenues, which lead to increased tax revenues for the government, and thus to increases in public spending. A number of the aforementioned elements are direct determinants of the size of the shadow economy in the general literature, and these constant cross-effects (i.e. each determinant leading to another determinant) can potentially build towards a reduced shadow economy.

Similarly, we see a similar, but possibly more relevant effect with import numbers. Interestingly enough, imports have been studied as less of a factor, yet theoretically they seem to be more important. This relevance comes from multiple dimensions. First of all, we can segment the many benefits of importing (as discussed by Sirgy et al. (2006)) and describe the first category as those that increase tax revenues. These are simply increases in jobs, increases in the sales of the importing sector, and transfers of technology which lead to increases in tax revenues, greater public services, and thus a reduced shadow economy. The other dimension of this analysis is that more imports lead to lower prices and greater access to high quality goods. Clearly lower prices in the formal economy will draw individuals there instead of the shadow economy. This is a simple demand theory of two "businesses",

the formal and informal businesses. It is dubious as to whether or not higher quality goods will lead people to engage in shadow economic activity. Consequently, we will also give a treatment of this relationship. To focus on purchases, we can see that Williams and Martinez-Perez (2014), in a large survey of European purchasers of informal goods, found that while most people had price in mind when purchasing goods in the shadow economy (this focuses on the purchase of legal goods, but without official reporting channels). A large number of purchasers cited inadequacy of goods in the formal sector as a reason for buying without reporting. The numbers are 64% and 38%, respectively. Thus, we can conclude that an improvement in the quality and price of goods imported (oftentimes coming directly from more imports) are likely to dampen the incentives for purchasing underground. The preceding analysis indicates that imports may be an important factor affecting shadow economic activity, even beyond the main reasons cited for larger shadow economies.

While there are clear reasons for exports and imports reducing the size of the shadow economy, there are also quite substantial reasons why it might lead to increases. To begin with exports, Sirgy et al. (2006) frame long-term export increases as conducive to national resentment of the exporting country by the importing one, this leads to a chain of events that, in the end, reduces many of the indicators discussed of the size of the shadow economy. Essentially an increase in exports, will later lead to a decrease as pushback for weakening domestic production. With the similarity to the short-run effects in result but contradiction in the direction, there is now ambiguity as to which direction increased exports should lead the shadow economy. Similarly, increases in imports can undo some of the effects that should have been present. The main avenue for this is in increased competition. Increased competition leads to greater unemployment and lower sales, which in turn reduces tax revenues and starts our cycle increasing shadow economic size again. Not only is this the case, but Tanzi (2002) and Schneider & Enste (2000) discuss competition as a driving force behind businesses moving to the shadow economy. In an attempt to compete with superior importers, they may try to reduce costs by hiring underground workers. They can also purchase more goods from the shadow market. We also have the small addition of decreased employment in importing sectors.

The preceding discussion reveals itself in the mixed results of similar studies thus far. A number of studies have varying levels of significance while all controlling for a different set of variables. To begin, Berdiev and Saunoris (2017) used Dreher’s 2006 and Dreher et al. ’s 2008 dataset of globalization (broken down into economic, political, and social). When using time and country fixed effects, the researchers find a negative, but statistically insignificant effect of economic globalization on two different estimates of the shadow economy (namely, Schneider et al. (2010) and Elgin and Öztunali (2012)).

To continue along similar lines, it is important to discuss the prevalence of tariffs. While tariffs act to reduce imports and may lead to reductions in exports as well with reduced business inputs quality and international retaliation, they are worth having a separate discussion. Luckily, theory regarding the shadow economy as well as empirical results tell us that taxes levied are likely to increase the size of the shadow economy. Tanzi (2002) also references trade-related taxes as indicators of more underground activity. The main focus of this analysis is to see whether or not the cross-effects and ambiguity of trade as a determinant of the shadow economy discussed earlier will cause tariffs to have a larger or smaller effect than the most commonly cited taxes. Oddly enough, Fiess and Fugazza (2012) find that tariffs have a negative effect on trade although the significance of these results vary depending on the econometrics model used and the dataset on informality.

3 Data and Variables

The quality of data used is clearly of utmost importance. The following utilizes the most reliable estimates available for the dependent variable and collects independent variables and controls from reputable sources such as the International Monetary Fund, the World Development Indicators data from the World Bank, the Heritage Foundation, and World Integrated Trade Solutions. Some of the variables used are scaled (e.g. per capita metrics) by using WDI's datasets where needed.

3.1 Dependent Variable

The dataset used for the dependent variables (the size of the shadow economy as a percentage of GDP) is the Multiple Indicators Multiple Causes (MIMIC) model estimations done by Schneider and Medina (2018). This represents the ideal dataset to use as Schneider is the lead expert on the topic of the shadow economy globally and the MIMIC model captures the effects of a variety of variables related to the shadow economy versus just a single one, mainly taxation, like other estimation methods (Schneider and Medina, 2018). There are a few limitations of the model such as a difficulty of excluding traditional crime, unstable coefficients when the model specifics or sample size is changes, and benchmarking proceduers are not very robust when obtaining "real world figures" (Schneider and Medina, 2018).

3.2 Independent Variables

Maintaing along the lines of the 3 pillars of trade defined at the start of section 2.3, we use the following variables.

We start with the taxes on international trade, called **Trade Taxes** in the model. This dataset encompasses all taxes levied on trade. The Australian Bureau of Statistics defines taxes on international trade as "taxes that become payable when goods cross the national or customs frontiers of the economic territory..." (2015). They further fragment this into customs duties on imports and exports, agricultural produce export taxes, and others. This variable is defined as trade taxes as a percentage of GDP and is used to capture the significance of the trade barriers put in place by these taxes. This data was taken from the IMF's World Revenue Longitudinal Database.

We continue on to a similar variable, the weighted average applied tariff rate of the countries (called **Applied Tariff Rate** in the model). This variable, as the name suggests, is the weighted average (all products) of applied tariffs levied by each country to their partner countries. This variable is similar to the other one, but captures not the relational monetary value collected from taxes, but what effect the current rates of taxation have on a countries shadow economic movements. This data was taken from the World Development Indicator's (WDI) "Tariff rate, applied, weighted mean, all products (%)" dataset. This variable and the former reprsent the trade barriers aspect of the overall definition, capturing major indicators that might induce changes in trading. These variables are also important because, as discussed earlier, taxation is a major driver of shadow economic activity. The attempt here is to understand whether or not tax-trade variables have an outsized effect on the shadow economy.

Moving onto the second pillar, we have both **imports per capita** and **exports per capita**. These variables are meant to capture relationships between trade volume per individual and the size of the shadow economy. Based on Sirgy et. al.'s (2006) analyses on trade

and quality of life, the volume of trade and how many products people have access and can profit off of seem like they may have an effect on the shadow economy, so they are included in the model. These variables were taken from WDI, Imports and Exports (constant prices at USD 2015), respectively and then divided through by each countries population for the year (population data was also taken from WDI).

And finally, we want to capture the quality of goods that the individuals in a country have access to pre-trade by taking the weighted average of their reveal comparative advantages across a wide variety of product classes (called **Weighted Average RCA**). The reveal comparative advantage (RCA) of a specific product can be defined by the formula...

$$RCA_{Ai} = \frac{\frac{X_{Ai}}{\sum_{j \in P} X_{Aj}}}{\frac{X_{wi}}{\sum_{j \in P} X_{wj}}}$$

Where...

- P is the set of all products (with $i \in P$),
- X_{Ai} is the country A's exports of product i,
- X_{wi} is the worlds's exports of product i,
- $\sum_{j \in P} X_{Aj}$ is the country A's total exports (of all products j in P), and
- $\sum_{j \in P} X_{wj}$ is the world's total exports (of all products j in P).

This formula simply represents the ratio of exports of a product versus total exports of a country relative to the same ratio in the world. The rational behind this variable is that if a country has an advantage in producing a variety of products, then individuals will be less willing to supplement purchases in the shadow economy. Both the RCA's of each product category and exports of the same product categories were taken from the World Integrated Trade Solutions database. It is important to note that the product classes chosen will have some overlap (this was seen when summing them together lead to values greater than export numbers) which can lead to imperfect weighted averages.

3.3 Control Variables

The main justification for the control variables is the discussions done by Schneider and Enste (2000) and Sneider and Buehn (2017). These are summarized in Figure 2.

3.4 A Note on Data Cleaning

Before moving onto the results, it is important to note the practices regarding missing data in the paper. Although infrequent, missing values were found across a number of the country and, if one were to retain only countries that had no missing values, there wouldn't be much of a dataset left. To account for this issue, a simple linear interpolation was used to fill in missing values between years.

4 Intermediate Results

To begin, we used the following simple linear model as a starting point for the analysis,

$$se_size_{it} = \sum_{x \in X} \beta_i x_{it} + \sum_{y \in Y} \gamma_i y_{it} \quad (1)$$

Variable	Description
Composite WGI	The Composite WGI is a simple average of the Worldwide Governance Indicators dataset components of <i>control of corruption</i> , <i>government effectiveness</i> , <i>regulatory quality</i> , and <i>rule of law</i> . They range from -2.5 to 2.5 and were scaled by 4. This variable is intended to describe the important determinants of the shadow economy relating to corruption and regulatory quality. Source: WDI - WGI Indicators
GDP per Capita	This is simply meant to control for income levels and is a commonly used control variable in similar studies. This was calculated by taking the GDP of each country and dividing it by the population. Source: WDI - GDP
Income Taxes	As this is another tax that is an important determinant of the shadow economy (generally with regards to trying to work in the underground economy to avoid paying this tax), we will include it as a control variable. This is measured as income taxes collected as a percentage of GDP. Source: IMF - World Revenue Longitudinal Database
Taxes on Goods and Services	Continuing along the lines of taxes, we will also include this calculation as a control variable. This is also measured as a percentage of GDP. Source: IMF - World Revenue Longitudinal Database
Tax Burden	This variable is a simple average of the top marginal tax rate on individual income, the top marginal tax rate on corporate income, and the total tax burden as a percentage of GDP (Index of Economic Freedom, 2026). The final factor is the sum of all direct and indirect taxes levied by the government (all levels), so there may be some collinearity between this variable and the other tax variables, but this variable is used to capture the effects of the marginal tax rates which are posited to help stop increases of the size of the shadow economy contingent on stopping increases in this tax rate (Cebula, 1997). Source: World Heritage Foundation - Index of Economic Freedom

Figure 2: Table of Control Variables

for X is the set of independent variables and Y is the set of control variables as defined in the methodology. We report the results of the initial-stage, simple models below. This begins with pooled OLS, random effects, and country-fixed-effects. The following doesn't include nonlinearities or interaction terms. This is done to mirror the current research on the topic to see if the following are more robust. To start, we can use a Hausman test to determine which model might present better results.

```
data: form
chisq = 62.243, df = 10, p-value = 0.00000000136
alternative hypothesis: one model is inconsistent
```

From this we deduce that the random effects model is will not have consistent estimators and, thus, is not very robust. Based on this, the results stated up to this point will be based on the results of the country-fixed-effects model in column (6) (Figure 5). Figure 3 also presents the correlation matrix.

What we can see from these results is that taxes have a significant effect on the size of the shadow economy. Both trade-tax variables (**Trade Taxes** and **Applied Tariff Rate**) are statistically significant beyond the 1% level. To focus in on **trade taxes** specifically,

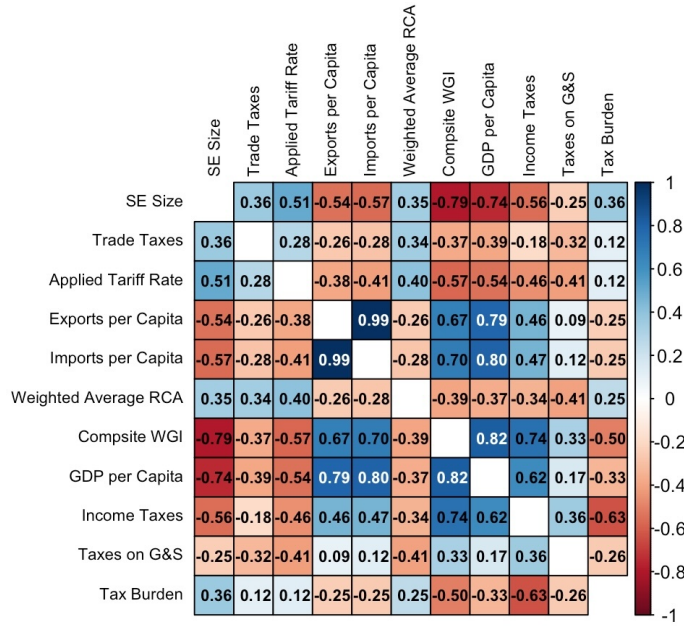


Figure 3: Correlation Matrix

Trade Tax	Applied Tariff Rate	Exports per Capita	Imports per Capita
1.3875	2.0323	64.9890	68.0874
Weighted Average RCA	Composite WGI	GDP per Capita	Income Tax
1.4347	5.6609	5.8336	3.2162
	Taxes on G&Ss	Tax Burden	
	1.5982	1.9970	

Figure 4: Variance Inflation Factors

we see that the relationship is negative. This runs contrary to what theory and results of taxation in general suggest. This variable captures customs duties and import/export taxes (as described by the IMF's metadata of its 'World Revenue Longitudinal Dataset'). Interesting enough, no research has been discovered that would suggest this relationship.

The other important tax variable is the effectively applied tariff rates (weighted average). This relationship being positive supports the theory that higher taxation leads to greater shadow economic activity. As tariff rates go up on average, people will be more inclined to participate in the underground economy. This is supported by Tanzi (2002) who stated that foreign trade taxes are one of the 6 major tax categories that influence the shadow economy.

An interesting result is that the levels of imports and exports do not have a significant effect on the size of the shadow economy. Both of these trends do, however, tend upwards. This is also contrary to the hypothesis that greater access to imports and a large exports count (for jobs) will reduce the size of the shadow economy. This relationship may persist because as more trade takes place, there becomes greater incentive to participate in the shadow economy.

A major result of the prior figures is the large multicollinearity present between exports per capita and imports per capita. This can go a way into explaining why the coefficient accompanying these variables are extremely low and why they fail to be statistically signif-

Figure 5: Contemporaneous regression results

	Pooled	Pooled	Random	Random	Fixed	Fixed
Trade Taxes	0.992*** (0.188)	0.231 (0.147)	-0.353 (0.253)	-0.682*** (0.177)	-0.438 (0.251)	-0.748*** (0.176)
Applied Tariff Rate	0.757*** (0.084)	0.108 (.007)	0.563*** (0.102)	0.356*** (0.073)	0.564*** (0.102)	0.354*** (.0737)
Exports Per Capita	0.00034* (0.00016)	0.000108** (0.000)	0.00022 (0.00013)	0.00002 (0.00010)	0.00022 (0.00012)	0.000015 (.000105)
Imports Per Capita	-0.00085*** (0.00019)	-0.000041 (0.000)	-0.00057*** (0.00017)	0.00012 (0.00011)	-0.00054** (0.00017)	0.000152 (.000117)
Weighted Average RCA	0.463*** (0.124)	0.0765 (0.107)	0.294 (0.180)	0.205 (0.156)	0.264 (0.184)	0.195 (.160)
Composite WGI		-0.8305*** (.0505)		-0.524*** (0.110)		-0.00027*** (.149)
GDP per Capita		-0.000288*** (.00003)		-0.00026*** (0.00007)		-0.00027*** (.000075)
Income Taxes		0.2293*** (.06619)		-0.391* (0.170)		-0.443* (.188)
Taxes on Goods and Services		0.00167 (.0527)		-0.405* (0.160)		-0.436* (0.172)
Tax Burden		0.0137 (.0204)		-0.067* (0.032)		-0.070* (.034)
Intercept	24.735*** (0.693)	31.266*** (2.031)	26.770*** (1.481)	43.776*** (2.738)		
Fixed Effects Included	No	No	No	No	Yes	Yes
R^2	0.442	0.668	0.273	0.437	0.266	0.421
Adj. R^2	0.440	0.666	0.270	0.432	0.214	0.377
Num. obs.	1234	1234	1234	1,234	1234	1234
Residual Std. Err.	9.703	7.495				
F-statistic	194.4	246.466	457.425	942.009	83.485	83.438

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

icant. Clearly with a practically perfect correlation and variance inflation factors that go way beyond the heuristic 10, more work will be done in subsequent models to evaluate this issue further. A quick joint test of significance shows that these variables are significant to the 15% level (the following table summarizes these results). Although this still is not a major development, there is a decent chance that some further specifications (transforms, different functional forms, interactions, etc.) can shed more light on this issue.

Figure 6: Imports and Exports per Capita Joint Significance Test

F_{stat}	15% sig.	10% sig.	5% sig.	1% sig.
1.931 (14.55%)	1.9901	2.3070	3.0032	4.6228

5 Next Steps

Despite some results so far, there are a number of things left to do. These include (in no particular order):

- Model changes like inclusion of interactions and nonlinearities.
- Additional variables / changes to variables (mainly depends on data availability) such as import quaity (independent variable); sactions (independent variable); size of agricultural sector (control variable); unemployment (control variable); transforms of some variables like GDP, imports, exports, etc. instead of per capita.
- Endogeneity and Instrumental variable approaches.
- Finish Literature Review